AI Image Generator

13. Change Log and Roadmap

Version 2.0
22 October 2026

Project owner: Frazile AI
Platform: NVIDIA GeForce RTX 3090, Windows 11, Python 3.12

Z-Image Turbo, Stable Diffusion XL and its best fine-tunes, FLUX.1, LoRAs and Real-ESRGAN, served locally with Gradio in the Frazile AI studio UI, with per-image GPU telemetry recorded in SQLite and a live Admin panel.

13. Change Log and Roadmap

13.1 Change log

Increment 1: command-line generator (21 Oct 2026)

- Created the project folder, virtual environment, and requirements.txt; installed torch 2.14.1+cu126, diffusers 0.41, transformers, accelerate.
- Wrote generate.py with prompt, negative, steps, guidance, size, seed, count and model arguments. First image: 1024x1024 in 9.7 s, peak VRAM 10.5 GB.

Increment 2: web interface

- Added app.py using Gradio 6.30 with a gallery and an info box; model loaded once at startup and kept resident. Added run-app.bat and README.

Increment 3: prompt-driven controls

- Removed style, size, steps and guidance controls. Added prompt_parser.py, the live "Will generate" line and a collapsible help table.

Increment 4: telemetry and history

- Added gpu_info.py and db.py with a 45-column SQLite table; per-image timing and GPU sampling; the History tab.

Increment 5: real 4K and 8K output

- Added upscaler.py (Real-ESRGAN x4plus, tiled, multi-pass); parser upscale words; eight upscale columns; 4K in 3.4 s, 8K in 8.5 s.

Documentation 1.0 and repository (22 Oct 2026, 09:17)

- Documentation set published as PDFs under Frazile AI ownership. Project pushed to github.com/Fzhacking/ai-image-gen (private).

Increment 6: model registry (22 Oct 2026, 10:09)

- Added models.py: SDXL base, Juggernaut XL v9, RealVisXL V5.0, DreamShaper XL v2 Turbo, Playground v2.5, Animagine XL 3.1 and FLUX.1 schnell; local checkpoints; LoRAs; compel weighting; per-model samplers and defaults; ModelManager with verified VRAM release.
- Parser: model and LoRA directives, explicit-flag tracking, apply_defaults(). App: Model dropdown with Auto, model in details and history. CLI: --model, --lora, --list-models.
- Installed peft 0.21, compel 2.5, bitsandbytes 0.50, sentencepiece, protobuf. Downloaded the five fine-tunes (32.5 GB).
- Fixed: compel padding for SDXL dual encoders; comma tidy-up; FLUX fidelity note.
- Measured: every SDXL-family model at 832x1216 in 2.0 to 10.5 s, peak 13.2 GB, swap 26 s.

Increment 7: Frazile AI branding and Upscale bar (22 Oct 2026, 10:27)

- Dark studio theme with pink-to-yellow accents, applied to both colour modes; branded header; Inter and JetBrains Mono.
- Upscale bar after generation: 2K / 4K / 8K, selected image or whole batch, records updated in place with the new db.update().

Increment 8: full width, Admin panel, Docker (22 Oct 2026, 10:50)

- Full-width layout, version pills, Clear button, "Loaded now" line.

- Admin tab with nine tiles and six sections; sysinfo.py collector on psutil and nvidia-smi; 5 s auto-refresh; optional reachability check; caching so a refresh costs 0.7 s.
- Dockerfile, docker-compose.yml and .dockerignore; app reads bind address, port and browser flag from the environment.

Increment 9: Z-Image Turbo as the default model (22 Oct 2026)

- Added the zimage family to models.py with a bf16 loader and CPU offload under 32 GB; Z-Image Turbo (Tongyi-MAI, 6B, 8 steps, Apache 2.0) is the model loaded at start-up.
- The repository ships the transformer in fp32 (23 GB); the first load converts it to bf16 and saves a 12 GB copy under models/cache/ so later loads take about about 130 s. The Qwen3 text encoder is loaded in 8-bit; everything stays on the GPU (15.8 GB resident, 18.1 GB peak).
- Measured: 1344x768 in 9.8 s and 832x1216 in 8.6 s at 8 steps; the FRAZILE AI neon-sign test rendered the text correctly.
- Auto no longer switches away from Z-Image or FLUX; style-based specialist selection applies only while an SDXL-family model is loaded.
- Fixed: CPU offload of Z-Image crashed the process on the 16 GB machine (fp32 shards plus offload exceeded system RAM); replaced by shard-by-shard loading straight onto the GPU.

Increment 10: starter LoRAs and Qwen-Image (22 Oct 2026)

- Pixel Art XL and Papercut LoRAs fetched by models.py fetch-loras; LoRAs box in the UI; with Auto, a requested LoRA switches to an SDXL model so it takes effect.
- Qwen-Image added: 20B transformer quantised to NF4 on first load and cached (11 GB), 4-bit text encoder from diffusers/qwen-image-nf4, Lightning 8-step LoRA as a permanent adapter with the exponential-shift scheduler.
- Measured: 13 to 14 s per 1 MP image at 8 steps, 18.0 GB peak, 90 s cached load. Posters, neon signs and menus render their text correctly.
- Fixed: VRAM paging on the second image (VAE tiling, cache release, expandable segments); FLUX gated-repo error now reported clearly.

Documentation 2.0 (22 Oct 2026)

- This document set, rebuilt from docs-src/build_pdf.py with three new sections (Models, Admin panel, Deployment) and a model comparison figure.

13.2 Roadmap

Priority	Item	Notes
High	Image-to-image and inpainting	Same pipeline family; lets a chosen image be edited rather than regenerated; the ModelManager already shares components
High	Character consistency with IP-Adapter or reference images	Keeps a subject's identity across prompts; the usual next step after LoRAs
High	Test the Docker image on a host with WSL 2 and Docker Desktop	Written and reviewed but unbuilt on the development machine
Medium	FLUX.1 dev and ControlNet	dev is gated and non-commercial; ControlNet gives pose and depth control
Medium	SDXL refiner or hires-fix pass	Adds generated detail at 2x before Real-ESRGAN
Medium	Export history to CSV and chart timings per model	pandas is already present
Medium	Thumbnail cache for the History grid	Avoids loading full PNGs when browsing

Priority	Item	Notes
Low	Authentication	Gradio supports basic auth; needed before exposing beyond localhost or publishing the container port
Low	Cloud deployment profile	The Docker image is the starting point for RunPod or a Hugging Face Space